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MIL-E-49149C(CR)
27 May 1992
SUPERSEDING
MIL-E-49149B(ER)
8 November 1989

MILITARY SPECIFICATION

ELECTRONIC CONTROL ASSEMBLY
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Electronic Control Assembly herein referred to as "electronic control", which is part of Viewer, Driver's Night Vision AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

SPECIFICATIONS

MILITARY

MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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STANDARDS

MILITARY

MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
DOD-STD-2000/4	-	General Purpose Soldering Requirements for Electrical and Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771336 - Electronic Control Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The electronic control is mounted to and limits operation of the Viewer, Driver's Night Vision AN/VVS-2(V) under damaging light brightness level.

3.2 Construction. The electronic control shall be constructed in accordance with SM-D-771336 and as specified herein.

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3.3 Qualification. Unless otherwise specified by the contract or purchase order, Electronic Control Assemblies furnished under this specification shall be products approved as qualified products (see 3.10.3 and 6.3).

3.3.1 Initial production testing. The supplier shall furnish and test electronic controls as specified in the contract or purchase order (see 6.2.d).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Power output. When the photo-detector is blocked, the electronic control shall supply +2.7 ± 0.2 volts direct current (Vdc) to pin 1 when the electronic control is energized with a BA-5567/U battery and +2.7 ± 0.2 Vdc when the electronic control is energized with the power adapter.

3.5.2 Power cutoff. The output voltage of the electronic control shall be reduced to 0.2 Vdc or less when the photodetector views a bright target area.

3.5.3 Power restoration. Output voltage shall return to within + 0.2 Vdc of the input voltage within 5 seconds after the brightness of the target area is reduced below the cutoff value of the photodetector. The input voltage shall be 3.0 Vdc.

3.6 Environmental characteristics.

3.6.1 High temperature. The electronic control shall operate at a temperature up to +52°C. The electronic control shall withstand storage temperature of +68°C without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.2 Low temperature. The electronic control shall operate at a temperature down to -51°C. The electronic control shall withstand storage temperature of -51°C without damage or degradation of performance (see 3.10.1 and 3.10.5)

3.6.3 Temperature shock. The electronic control shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.4 Vibration. The electronic control shall withstand vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation without damage or degradation of performance (see 3.10.1 and 3.10.5).

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3.6.5 Shock.

3.6.5.1 Basic shock. The electronic control shall withstand a basic shock of 40 g's (see 3.10.6) for a duration of 18 milliseconds without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.5.2 Weapon shock. The electronic control shall withstand simulated weapon firing shock of 100 g's (see 3.10.6) at 1.5 milliseconds on half sine pulse without damage or degradation of performance (see 3.10.1 and 3.10.5).

3.6.6 Burn-in/environmental stress screening (ESS). Each electronic control shall undergo 48 hours of light level cycle operation at 55°, ± 5°C. At the end of this cycle operation there shall be no evidence of damage (see 3.10.1).

3.7 Marking. Unless otherwise specified, electronic control components shall be marked and identified per MIL-M-13231.

3.8 Treatment and painting. Unless otherwise specified, treatment and painting shall conform to MIL-STD-171.

3.9 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any reference specifications or drawings where inspected in accordance with Section 4. Parts and assemblies shall comply with requirement 9 of MIL-STD-454 and soldering shall comply with DOD-STD-2000/4.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as 23°C, ± 10°C.

3.10.3 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.10.4 Environmental tests. For the purpose of this specification environmental tests are those tests discussed in 4.6.

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3.10.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.10.6. "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may be of use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspection shall be as classified as follows:

a. Initial production tests (4.3).
b. Quality conformance inspection (4.4).
c. Inspection for packaging (4.7).

4.3 Initial production testing.

4.3.1 Inspection. Unless otherwise specified (see 6.2.d), ten electronic controls manufactured with production tooling shall be selected from the first production lot and provided for IPT. Each IPT electronic control shall be inspected in accordance with Table I, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that electronic control and may be cause for rejection of IPT.

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TABLE I. Inspection.

Inspection	Requirement paragraph
Any part or materials not as specified	3.2, 3.4
Treatment and painting not as specified	3.8
Marking and painting not as specified or illegible	3.7
Workmanship	3.9

4.3.2 IPT Test. Following successful completion of the inspections specified in 4.3.1, each IPT electronic controls shall be subjected to all the tests in Table II. After Burn-In/ESS the balance of the tests may be performed in any order. Failure of any test shall be cause for rejection of the IPT.

4.3.2.1 Disposition of IPT samples. Disposition of IPT samples shall be in accordance with the contract or purchase order (see 6.2.d).

TABLE II. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Burn-in/environmental stress screening (ESS)	3.6.6	4.6.6
Power output	3.5.1	4.5.1
Power cutoff	3.5.2	4.5.2
Power restoration	3.5.3	4.5.3
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.4 Quality conformance inspection.

4.4.1 Examination. Each electronic control shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Test.

4.4.2.1 Group A. Each electronic control shall be tested in accordance with Table III with Burn-In/ESS being conducted first. Failure of any test shall be cause for rejection of that unit. Electronic controls which pass Group A inspection shall be formed into lots by the contractor.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Burn-In/ESS	3.6.6	4.6.6
Power output	3.5.1	4.5.1

4.4.2.2 Group B. Group B tests shall be conducted on electronic controls selected from lots which have passed the examination and the Group A test. Sampling shall be in accordance with the contract or purchase order. Tests shall be conducted in accordance with Table IV (see 6.2.c).

TABLE IV. Group B test schedule.

Inspection	Requirement paragraph	Test paragraph
Power cutoff	3.5.2	4.5.2
Power restoration	3.5.3	4.5.3

4.4.2.3 Group C. This inspection shall consist of the tests specified in Table V, and shall be performed on samples from lots that have been subjected to and have met Group A inspection. Sampling shall be in accordance with the contract or purchase order (see 6.2.c).

4.4.2.3.1 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to Group C tests shall be refurbished, pass the requirements of Group A tests and shipped against the requirements of the contract.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2.b).

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4.5 Performance Test procedure. Unless otherwise specified all test shall be performed at ambient temperature (see 3.10.2).

4.5.1 Power output. Energize the electronic control with the power adapter or equivalent power source. A 51, ± 2 ohm load shall be placed between pins 1 and 4. With the photodetector covered, measure the voltage between pins 1 and 4. The preceding measurement shall be repeated with the control energized with a BA-5567/U battery. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Power cutoff. Using the procedure of 4.5.1, measure the voltage after the photodetector is uncovered and exposed to a target of brightness between .1 footlambert and .18 footlamberts. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Power restoration. After performing the test of 4.5.2, reduce the brightness of the target area to less than .05 footlambert. Failure to meet the requirement of 3.5.3 shall constitute failure of this test.

4.6 Environmental. The electronic control shall be retested to meet the requirements of 3.5.1, 3.5.2, and 3.5.3 whenever an operating test is required during an environmental test, and at the completion of each environmental test (see 3.10.4), except for burn-in/environmental stress screening. Failure to meet these requirements shall constitute failure as defined in degradation of performance (see 3.10.5). The temperature of chambers shall be controlled to ± 2.0°C environmental test shall be as defined in 3.10.4.

4.6.1 High temperature. The electronic control shall be subjected to the test of Method 501.1, Procedure I, MIL-STD-810, except that the temperature of Steps 2 and 3 (highest storage temperature) shall be +68°C and the temperature of Steps 4 and 5 (highest operating temperature) shall be +52°C. Steps 4 and 6 stabilization period shall be a minimum of two hours, and Step 5 operating test shall conform to 4.6. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The electronic control shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -51°C. Low operating temperature (Step 4) shall be -51°C. Steps 2, 4, and 6 stabilization period shall be a minimum of two hours, and Step 5 operating test shall conform to 4.6. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.2 shall constitute failure of this test.

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4.6.3 Temperature shock. The electronic control shall be subjected to the test of Method 503.1, Procedure I of MIL-STD-810, except the high temperature chamber shall be maintained at +68 °C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.3 shall constitute failure of this test.

4.6.4 Vibration. The electronic control shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, Tracked Vehicles, limited to one sweep in the vertical axis only. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.4 shall constitute failure of this test.

4.6.5 Shock. The electronic control shall be subjected to the test of MIL-STD-810, Method 516.2.

a. 40 g's at a duration of 18 milliseconds per Figure 516.2-1 shall be the test, Shock Procedure I.

b. Weapon firings, 100g's using procedure IV except at 1.5 millisecond (ms), per Figure 516.1-2.

Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.5 shall constitute failure of this test.

4.6.6 Burn-in/environmental stress screening. With operating voltage applied, each electronic control shall be subjected to a temperature of +55 degrees ± 5°C. Once each 15 minutes during exposure at .20 footlambert, the electronic control shall be illuminated with less than .1 footlambert. To verify proper operation, hold the brightness to .20 fL while observing the electronic control's function as per 3.5.1. Failure to meet the requirements of 3.6.6 and 3.5.1 shall constitute failure of this test.

4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electronic control is used with the AN/VVS-2(V) for driver's viewing during periods of darkness.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Necessary actions by contractor in the event of a Group C failure (see 4.4.2.3.2).
c. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
d. Quantity and schedule for IPT testing for all line items and disposition of IPT samples.
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
f. MIL-STD-810C is used for environmental tests of section 4.
g. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.10.

6.5 Subject term keyword listing.
Electronic Control
Viewer Driver's

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-49149C(CR)	2. DOCUMENT DATE (YYMMDD) 27 May 1992

3. DOCUMENT TITLE
ELECTRONIC CONTROL ASSEMBLY VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME CECOM	b. TELEPHONE (Include Area Code) (1) Commercial (908) 532-5851	(2) AUTOVON 992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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